

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA0304	Course Title:	Mobile computing
CO Assessed:	CO1	Assessment:	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	Total Marks:	100 Marks
CO1 Statement:	<i>Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.</i>		
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Section / Batch:	CSE-AI	Date:	08-10-2026

Code of Conduct

I HASTHAVARAM SANJAY REDDY /192372056(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: HASTHAVARAM SANJAY REDDY

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department

Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

Q1. A mobile user streams a 3-minute video clip encoded at 1.5 Mbps. How much data, in MB, does the clip consume?

Answer:

Data = bit rate $\times$ time

1.5 Mbps $\times$ (3 $\times$ 60 s) = 270 Mb

Result: 270 Mb $\div$ 8 = 33.75 MB

Q2. A subscriber has a 30 GB monthly data plan and uses 1.2 GB every day. For how many days will the plan last?

Answer:

Days = plan $\div$ daily usage

30 GB $\div$ 1.2 GB

Result: 25 days

Q3. A smartphone battery is rated 5000 mAh at a nominal 3.85 V. How much energy, in watt hours, does it store?

Answer:

Energy (Wh) = capacity (Ah) $\times$ voltage

5 Ah $\times$ 3.85 V

Result: 19.25 Wh

Q4. A messaging app sends a 2 KB keep-alive message every 30 seconds. How much data does it use in one day?

Answer:

Messages per day = 86400 $\div$ 30 = 2880

Data = 2880 $\times$ 2 KB = 5760 KB

Result: $\approx$ 5.76 MB per day

Q5. Co-channel cells are located with shift parameters $i = 3$ and $j = 2$. What is the resulting cluster size N ?

Answer:

$N = i^2 + ij + j^2$

$3^2 + (3 \times 2) + 2^2 = 9 + 6 + 4$

Result: $N = 19$

Q6. Two co-channel cells of radius 2 km are 13.86 km apart. What cluster size does this reuse distance indicate?

Answer:

$D/R = \sqrt{3N}$

$$13.86 / 2 = 6.93 \rightarrow 3N = 6.93^2 \approx 48$$

Result: N = 16

Q7. Find the area covered by one hexagonal cell of radius 3 km.

Answer:

$$\text{Area} = (3\sqrt{3}/2) R^2$$

$$2.598 \times 3^2$$

Result: $\approx 23.38 \text{ km}^2$

Q8. A system has 840 duplex channels in total and uses a 12-cell cluster. How many channels does each cell get, and what is the frequency reuse factor?

Answer:

$$\text{Channels per cell} = 840 \div 12 = 70$$

$$\text{Reuse factor} = 1/N$$

Result: 1/12

Q9. A system with 336 channels and a 7-cell cluster covers its service area with 49 cells. How many calls can it carry at the same time?

Answer:

$$\text{Channels per cell} = 336 \div 7 = 48$$

$$\text{Total calls} = 48 \times 49 \text{ cells}$$

Result: 2352 simultaneous calls

Q10. A cell of radius 4 km is split into cells of radius 2 km in an area with path-loss exponent 4. By how many dB must the transmit power of the new cells be reduced?

Answer:

$$\Delta P = 10 \cdot n \cdot \log(R1/R2)$$

$$10 \times 4 \times \log(4/2) = 40 \times 0.301$$

Result: $\approx 12 \text{ dB}$

Q11. A cell with 84 channels is divided into six 60° sectors with channels shared equally. How many channels does each sector receive?

Answer:

$$\text{Channels per sector} = \text{total} \div \text{sectors}$$

$$84 \div 6$$

Result: 14

Q12. A city of 400 sq. km is served by cells of 8 sq. km each, grouped into 5-cell clusters. How many clusters are needed?

Answer:

$$\text{Cells needed} = 400 \div 8 = 50$$

$$\text{Clusters} = 50 \div 5$$

Result: 10 clusters

Q13. A 12-cell reuse plan operates with path-loss exponent 4 and six first-tier interferers. Estimate the S/I in dB.

Answer:

$$q = \sqrt{3N} = \sqrt{36} = 6$$

$$S/I = q^n / 6 = 6^4 / 6 = 216$$

Result: $10 \log(216) \approx 23.3 \text{ dB}$

Q14. A 7-cell system operates in an area with path-loss exponent 3 and six first-tier interferers. What S/I does it achieve?

Answer:

$$q = \sqrt[3]{(3 \times 7)} = 4.583$$

$$S/I = q^3 / 6 = 96.25 / 6 = 16.04$$

Result: $10 \log(16.04) \approx 12 \text{ dB}$

Q15. A receiver needs a C/I of at least 15 dB, and the serving signal is -75 dBm. What is the highest interference level it can tolerate?

Answer:

$$C/I = \text{Signal} - \text{Interference (in dB)}$$

$$I = -75 - 15$$

Result: -90 dBm

Q16. Find the length of a half-wave dipole antenna for a 900 MHz handset.

Answer:

$$\lambda = c/f = 3 \times 10^8 / 900 \times 10^6 = 0.333 \text{ m}$$

$$\text{Half-wave dipole} = \lambda/2$$

Result: $\approx 0.167 \text{ m (16.7 cm)}$

Q17. Find the free-space path loss over 100 m at 2.4 GHz.

Answer:

$$\text{FSPL} = 20 \log(d \text{ km}) + 20 \log(f \text{ MHz}) + 32.44$$

$$20 \log(0.1) + 20 \log(2400) + 32.44 = -20 + 67.6 + 32.44$$

Result: $\approx 80 \text{ dB}$

Q18. A base station transmits at 43 dBm with a 15 dBi antenna. The handset antenna gain is 0 dBi and the path loss is 130 dB. What power does the handset receive?

Answer:

$$P_r = P_t + G_t + G_r - \text{Path loss}$$

$$43 + 15 + 0 - 130$$

Result: -72 dBm

Q19. A car moves at 60 km/h while using a 2600 MHz carrier. Find the maximum Doppler shift.

Answer:

$$v = 60 \text{ km/h} = 16.67 \text{ m/s}$$

$$f_d = v \cdot f / c$$

Result: $16.67 \times 2.6 \times 10^9 / 3 \times 10^8 \approx 144.4 \text{ Hz}$

Q20. Find the thermal noise floor of a 200 kHz GSM channel at 290 K.

Answer:

$$N = -174 \text{ dBm} + 10 \log(B)$$

$$-174 + 10 \log(200000) = -174 + 53$$

Result: $\approx -121 \text{ dBm}$

Q21. A GSM TDMA frame lasts 4.615 ms, and a user is given one slot in every frame. How many slots does that user get per second?

Answer:

Slots per second = $1 \div \text{frame duration}$

$1 / 0.004615$

Result: ≈ 216.7 slots/s

Q22. A GSM control multiframe contains 51 TDMA frames of 4.615 ms each. How long does one multiframe last?

Answer:

Duration = frames $\times$ frame time

51×4.615 ms

Result: ≈ 235.4 ms

Q23. A CDMA link has a processing gain of 21 dB, and the receiver needs an E_b/N_0 of 7 dB. What is the jamming (interference) margin, ignoring system losses?

Answer:

Jamming margin = $G_p - E_b/N_0$

$21 \text{ dB} - 7 \text{ dB}$

Result: 14 dB

Q24. An OFDMA system uses 15 kHz subcarrier spacing. What is the useful OFDM symbol duration?

Answer:

$T_u = 1 / \Delta f$

$1 / 15 \text{ kHz}$

Result: $\approx 66.7 \mu\text{s}$

Q25. Handsets use pure (unslotted) ALOHA on a random access channel with offered load $G = 0.5$. What throughput is achieved?

Answer:

Pure ALOHA: $S = G \cdot e^{(-2G)}$

$0.5 \times e^{(-1)}$

Result: ≈ 0.184

Q26. A slotted random access channel has 1000 slots per second, and 200 access attempts arrive each second. How many attempts succeed per second?

Answer:

$G = 200 / 1000 = 0.2$ per slot

$S = G \cdot e^{(-G)} = 0.2 \times 0.8187 = 0.164$

Result: Successes = $0.164 \times 1000 \approx 164$ per second

Q27. A paging channel offers 40 paging blocks per second, and each block carries 2 pages. A location area needs 50 pages per second. What is the paging channel utilisation?

Answer:

Capacity = $40 \text{ blocks} \times 2 \text{ pages} = 80 \text{ pages/s}$

Utilisation = demand $\div$ capacity = $50 / 80$

Result: 62.5%

Q28. A BTS has 4 TRXs with 8 time slots each, and 2 slots are reserved for signalling. How

many traffic channels are available?

Answer:

$$\text{Total slots} = 4 \times 8 = 32$$

$$\text{Traffic channels} = 32 - 2$$

Result: 30

Q29. A BSC can handle 512 TRXs. The network has 400 BTSs with 3 TRXs each. How many BSCs are needed?

Answer:

$$\text{Total TRXs} = 400 \times 3 = 1200$$

$$\text{BSCs} = 1200 \div 512 = 2.34$$

Result: Round up = 3 BSCs

Q30. An E1 link has 32 time slots of 64 kbps, of which 2 are used for synchronisation and signalling. Each remaining slot carries four 16 kbps traffic channels. How many traffic channels does the E1 carry?

Answer:

$$\text{Usable_slots} = 32 - 2 = 30$$

$$\text{Channels} = 30 \times 4$$

Result: 120 traffic channels

Q31. An EIR stores an 8-byte record for each registered IMEI. How much storage is needed for 20 million devices?

Answer:

$$\text{Storage} = \text{record size} \times \text{devices}$$

$$8 \text{ bytes} \times 20,000,000$$

Result: 160 MB

Q32. A VLR requests 5 authentication triplets from the AuC for each of 100,000 subscribers per hour. How many triplets must the AuC generate per second?

Answer:

$$\text{Triplets per hour} = 100,000 \times 5 = 500,000$$

$$\text{Per second} = 500,000 \div 3600$$

Result: ≈ 139 triplets/s

Q33. A TRX carries 8 full-rate voice channels. If it switches to half-rate coding, how many users can it carry, and what is the capacity gain?

Answer:

Half-rate: each channel carries 2 users

$$\text{Users} = 8 \times 2 = 16$$

Result: Gain = $2 \times$ (100% increase)

Q34. A TCP segment carries 536 bytes of data with 20-byte TCP and 20-byte IP headers. What is the transmission efficiency?

Answer:

$$\text{Efficiency} = \text{data} \div (\text{data} + \text{headers})$$

$$536 / (536 + 20 + 20) = 536 / 576$$

Result: $\approx 93.1\%$

Q35. A 1500-byte IP packet is segmented into radio link blocks that each carry 40 bytes of

payload. How many blocks are needed?

Answer:

Blocks = packet size ÷ payload per block

$$1500 / 40 = 37.5$$

Result: Round up = 38 blocks

Q36. A radio link has a 10% block error rate, and every failed block is retransmitted until it succeeds. What is the average number of transmissions per block?

Answer:

Average transmissions = $1 / (1 - \text{BLER})$

$$1 / (1 - 0.1)$$

Result: ≈ 1.11

Q37. An SMS carries a 140-byte payload using 7-bit character coding. How many characters can one SMS hold?

Answer:

$$\text{Total bits} = 140 \times 8 = 1120$$

$$\text{Characters} = 1120 \div 7$$

Result: 160

Q38. A PLMN uses a 16-bit Location Area Code. What is the maximum number of location areas it can define?

Answer:

Number of LAs = 2^{bits}

$$2^{16}$$

Result: 65,536

Q39. A Cell Identity field is 16 bits long within each location area, and the LAC is also 16 bits. How many unique cells can a PLMN identify?

Answer:

Unique cells = $2^{(\text{LAC bits} + \text{CI bits})}$

$$2^{16} \times 2^{16} = 2^{32}$$

Result: 4,294,967,296

Q40. An SS7 signalling link runs at 64 kbps and is engineered for 40% load. If each MAP message is 150 bytes, how many messages can the link carry per second?

Answer:

$$\text{Usable rate} = 0.4 \times 64000 = 25,600 \text{ bps}$$

$$\text{Message size} = 150 \times 8 = 1200 \text{ bits}$$

Result: Messages/s = $25,600 \div 1200 \approx 21$

Q41. A data packet crosses 4 core-network nodes. Each node adds 2 ms of processing, and each hop adds 1 ms of transmission. What is the one-way core delay?

Answer:

$$\text{Per-node delay} = 2 \text{ ms} + 1 \text{ ms} = 3 \text{ ms}$$

$$\text{Total} = 4 \times 3 \text{ ms}$$

Result: 12 ms

Q42. An average call lasts 120 seconds, and a user moves into a new cell every 60 seconds. How many handoffs occur during an average call?

Answer:

Handoffs = call duration $\div$ cell dwell time

120 / 60

Result: 2

Q43. A pedestrian walks at 1.5 m/s through a microcell 500 m across. How long does the user stay in the cell?

Answer:

Time = distance $\div$ speed

500 / 1.5

Result: ≈ 333 s (≈ 5.6 min)

Q44. Out of 10,000 handoff attempts, 9700 succeed. What are the handoff success rate and the handoff failure rate?

Answer:

Success rate = $9700 / 10,000 \times 100 = 97\%$

Failure rate = $100\% - 97\%$

Result: 3%

Q45. Of 50 handoffs recorded on a road, 12 returned to the original cell within 5 seconds. What is the ping-pong handoff rate?

Answer:

Ping-pong rate = ping-pong handoffs $\div$ total handoffs

12 / 50

Result: 24%

Q46. A visited MSC has a pool of 5000 roaming numbers (MSRNs). Each is held for 30 seconds during call setup. How many roaming call setups can it handle per hour?

Answer:

Setups per MSRN per hour = $3600 \div 30 = 120$

Total = 5000×120

Result: 600,000 per hour

Q47. A location area has 10 cells. A user receives 6 calls per hour (each paged in all 10 cells) and makes 2 location updates per hour, each costing 5 messages. What is the total signalling per hour for this user?

Answer:

Paging = $6 \text{ calls} \times 10 \text{ cells} = 60 \text{ messages}$

Location_updates = $2 \times 5 = 10 \text{ messages}$

Result: Total = $60 + 10 = 70$ messages/hour

Q48. A cell receives 300 calls per hour with an average holding time of 3 minutes. What is the offered traffic in Erlangs?

Answer:

A = calls per hour $\times$ holding time (hours)

$300 \times (3/60)$

Result: 15 Erlangs

Q49. Of 50,000 call attempts, 48,500 are set up successfully. What is the call setup success rate?

Answer:

Success rate = successful $\div$ attempts $\times$ 100

48,500 / 50,000 $\times$ 100

Result: 97%

Q50. A cellular network is out of service for 26.28 hours in a year of 8760 hours. What is its availability?

Answer:

Availability = (total – downtime) $\div$ total

(8760 – 26.28) / 8760

Result: $\approx$ 99.7%